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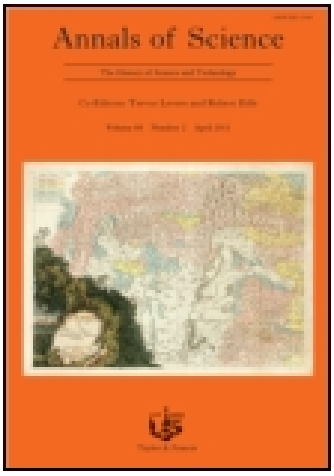

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The scholastic pendulum

Bert S. Hall [a]

[a] Institute for the History and Philosophy of Science and Technology, Centre for Medieval Studies, University of Toronto , Toronto, Canada , M5S 1A1

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The Scholastic Pendulum

BERT S. HALL
Institute for the History and Philosophy of Science and Technology,
Centre for Medieval Studies, University of Toronto, Toronto, Canada M5S 1A1

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Summary

The history of the physics of pendular motion rightly begins with Galileo's discovery of the isochronous character of that motion. There is, however, a ' pre-history ' of the pendulum, centering on its initial recognition as a significant special case requiring explanation. This occurred in the writings of Jean Buridan and Nicole Oresme in the middle of the fourteenth century. Earlier works that might have been construed as discussing pendular motion are considered, as are the explanations for the scholastic ' discovery ' of pendular motion put forth by Thomas Kuhn and Piero Ariotti. In contrast to these writers, this paper seeks to account for the pendulum's emergence with reference to an imaginary experiment concerning a body moving past the earth's center, medieval theories of *impetus*, and the proximate physical model of pendular motion, the late medieval heavy suspended church bell.

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1. Introduction

In the history of physics the pendulum plays a role of singular importance. From the early years of the seventeenth century, when Galileo announced his formulation of the laws governing pendular motion, to the early years of this century, when it was displaced by devices of superior accuracy, the pendulum was either an object of study or a means to study questions in astronomy, gravitation and mechanics. One of the more charming tales in the history of science concerns Galileo's supposed accidental discovery of the pendulum's regularity through observation of a swinging sanctuary lamp in the cathedral at Pisa (about 1582); since the story was told only after Galileo's death and lacks confirming evidence, a few historians take it seriously.[1] We find more

[1] Vincenzo Viviani, ' Racconto istorico della vita di Galileo ', in Galileo Galilei, *Opere*, vol. 19 (Edizione Nazionale; Firenze, Barbeia, 1907), 603. This version from 1654 mentions the lamp incident and a later pulse-timing instrument supposedly invented by Galileo only briefly. An expanded version appeared in a letter of 20 August 1659 to Prince Leopold de'Medici written to defend Galileo's priority over Christian Huygens in the invention of a pendulum-regulated clock (*ibid.*, 648). Viviani mentioned the incident once more in an unpublished treatise discussed by Silvio A. Bedini, ' Galileo Galilei and the measure of time ', in *Saggi su Galileo Galilei* (Firenze: Barbia, 1967), 35. There can be no doubt that Viviani sincerely believed the tale he told, but its accuracy is nevertheless highly questionable.

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interest in Galileo's early reports of his experiments with pendula as they appear in his correspondence from late 1602 onwards; these have been discussed from several different viewpoints in recent contributions to the literature.[2] Part of the story has not been explored in any depth, however, and that is the pre-Galilean concept of the pendulum which was first developed by scholastic philosophers in the fourteenth century. Though Galileo rendered this conception obsolete by his more intensive analysis of pendular motion, scholastic ideas held sway until his day. Their genesis and development is the subject of this paper, as is an attempt to deal with the philosophical problem they present to the historian, a problem of the relationship between ' seeing ' and ' knowing '.

In good scholastic fashion, let us begin with a few distinctions. First, Galileo's thesis of regularity in the pendulum's motion, technically known as ' isochrony ', and the pendulum itself, are not the same conceptual entity. This may seem too obvious to need much emphasis, but the two have not always been kept distinct in the literature. Pendular isochrony is Galileo's discovery, and nothing in this work is intended to deny him credit for it. Second, the pendulum as a device is too simple to be traced through time by the historian's conventional means. Surely the pendulum was ' invented ' by the first cave-man (or cave-child) who tied a piece of sinew rope to a rock and let the rock swing from his hand. Likewise, pendular motion is inherent in several devices common in Graeco–Roman antiquity, including the oscillations of plumb-bobs, swings and suspended battering-rams.[3] Our inquiry is concerned with the emergence of an idea in mechanics, with the process whereby the pendulum came to be recognized as a special case of motion demanding special explanations. That, as we shall see is far more complex than the construction of a simple pendulum.

[2] The letter in which Galileo announces his discovery was written to Guidobaldo del Monte on 29 November 1602 (*Opere* Ed. Naz., vol. 10 (1900), 97–100). An anti-empirical account of Galileo's discovery is found in Alexandre Koyré, ' An experiment in measurement ', *Proceedings of the American Philosophical Society*, **98** (1953), 224 *ff*. An analysis of the mathematics of Galileo's reasoning on this matter is given by Stillman Drake, ' Mathematics and discovery in Galileo's physics ', *Historia mathematica*, **1** (1974), 129–150. Thomas S. Kuhn, *The structure of scientific revolutions* (2nd ed.; Chicago: University of Chicago Press, 1970), 118 *ff*. presents a different interpretation. James MacLachlan, ' Galileo's experiments with pendulums: real and imaginary ', *Annals of science*, **33** (1976), 173–185, touches on the problem of discovery of isochrony. A somewhat broader context is dealt with by Piero Ariotti, ' Galileo on the isochrony of the pendulum ', *Isis*, **59** (1968), 414–426; 'Aspects of the conception and development of the pendulum in the 17th century ', *Archive for history of exact sciences*, **8** (1972), 329–410; and ' Toward absolute time: the undermining and refutation of the Aristotelian conception of time in the 16th and 17th centuries ', *Annals of science*, **30** (1973), 31–50. Compare also Lynn White, jr., ' Pumps and pendula: Galileo and technology ', in Carlo Golino (ed.), *Galileo reappraised* (Berkeley and Los Angeles: University of California Press, 1966), 108 *ff*.; and Ladislao Reti, ' Il moto dei proietti e del pendolo secondo Leonardo e Galileo ', *Le machine*, **1** (1968), 3–29. I would also like to express my gratitude to Professors White and Ariotti, Mr. Bedini, and the late Dr. Reti for having discussed the subjects of this article with me and for offering their advice and encouragement.

[3] The common plumb-bob becomes a pendulum when allowed to oscillate before coming to rest. It can be traced back to Egyptian antiquity (Charles Singer *et al.* (eds.), *A history of technology*, vol. 2 (Oxford: Clarendon Press, 1956), 386 and 672). The Graeco-Roman swing was usually a chair or stool suspended from four ropes (Franz Maria Feldhaus, *Die Technik der Vorzeit, der geschichtlichen Zeit und der Naturvölker* (Leipzig: 1914; reprinted München: Heinz Moos, 1965), col. 904). Such swings are probably mentioned by Hippocrates; compare H. G. Liddell and T. R. Scott, *A Greek–English lexicon* (new ed.; Oxford: Clarendon Press, 1940), *q.v.* αἰωρέω. The suspended battering-ram had been developed by the 5th century B.C. at the latest (Singer, vol. 2, 715–716).

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2. Antiquity and early middle ages: negative judgments

The most profoundly puzzling aspect of the pendulum's history is the slowness with which it came into the field of scientific thought and technological use. Despite its apparently simple character, the pendulum makes no appearance in Western thought before the middle of the fourteenth century, when it was discussed in different fashions in the works of two Parisian scholastics, Jean Buridan and Nicole Oresme. We shall discuss the texts and their contents shortly, but first it is necessary to review the evidence from earlier periods to see if it fully sustains the negative judgment. Perforce, any such argument must suffer from the logical difficulties of an argument from silence.[4] Still, I have been able to find no specific reference to pendular motion in any classical scientific or literary text, be it Aristotelian, Platonic or Stoic, nor is there the slightest indication that such a motion was ever glimpsed, but not discussed.[5] Moreover, no ancient machine seems to have been designed to make use of any property of the pendulum, though some ancient devices did display pendular motion as an accidental feature of their operation. Thus, not only did the pendulum's isochrony remain unknown before the age of Galileo, its very existence may be said to have lain unrecognized before the age of Dante.

At first glance such an assertion appears paradoxical, if not entirely incredible. We have already noted that the pendulum must have been 'invented' fairly early in man's experience, and that some ancient devices did display pendular motion. Surely with such examples before them, figures of the stature of Aristotle or Archimedes must have recognized at least some of the pendulum's peculiar properties. There is just no evidence, however, that they or any of their contemporaries did. If not in the West, then perhaps in China or Islam scholars and technicians found in the pendulum a subject for speculation and application? Again the answer is largely negative, with two minor qualifications.

[4] The suddenness with which the pendulum emerged in the fourteenth century seems first to have been noted by Kuhn (footnote 2), 120 *ff*. Ariotti, 'Development of pendulum' (footnote 2), 339 also credits Oresme with making 'the first direct reference to the pendulum'. We shall discuss the arguments advanced by both men in section 4 below. Lynn White, jr., *Medieval technology and social change* (London: Oxford University Press, 1962), 117, touches on the problem when he assigns credit to Leonardo da Vinci for the origins of pendulum-using machinery.

[5] Any argument from silence has an inexorable tendency to become an argument about the omniscience of the author. No claim is implied in this case that the author has a full and complete knowledge of everything written anywhere on the planet between 3000 B.C. and 1300 A.D. In seeking to verify the assertions of Kuhn, Ariotti and White noted above, I have sought to make a reasonably comprehensive survey of the works of Plato, Aristotle and the Stoics. I have relied on Samuel Sambursky, *The physical world of the Greeks* (New York: MacMillan, 1956), and *The physics of the Stoics* (New York: MacMillan, 1959), and Marshall Clagett, *Greek science in antiquity* (New York: Collier, 1955) as general guides to classical antiquity. Seyyed Hossein Nasr, *Science and civilization in Islam* (Cambridge, Mass.: Harvard University Press, 1968) was helpful for the Muslim world. George Sarton, *Introduction to the History of Science* (3 vols.; Baltimore: Williams and Wilkins, 1927–1947) has been extensively consulted, as has Joseph Needham *et al.*, *Science and civilisation in China* (5 vols. in 8 parts to date; Cambridge: Cambridge University Press, 1951–1976). For technology in both antiquity and the middle ages, I have relied on Charles Singer *et al.*, *A history of technology* (5 vols.; Oxford: Clarendon Press, 1954–1958) and Maurice Daumas (ed.), *A history of technology and invention* (2 vols.; New York: Crown, 1969). Various monographic studies will appear elsewhere in the footnotes to this article.

I might add that I began work on this phase of the research in a skeptical mood *vis-à-vis* Kuhn's claims (I had not then read Ariotti) and only gradually after much fruitless searching became convinced that the pendulum was entirely absent from classical and early medieval writings. At the very least, the onus is now on those who might wish to assert the contrary.

In China the only suitable candidate for a pendulum-using device might be the 'earthquake weathercock' invented in 132 A.D. by Chang Hêng 'Astronomer Royal' at the Imperial Bureau of Astronomy and Calendar in Peking.[6] Chang Hêng's instrument was meant to indicate the occurrence of earthquakes too small to be felt in Peking and to delineate the lines along which the shock waves were directed. The central mechanism of the 'seismometer' was a delicately balanced column;[7] when set in motion by a shock wave, it tripped one of several indicators that surrounded the column. The act of tripping the indicator brought into play a latch mechanism that froze the column to prevent any further motion.[8] Chang Hêng's machine is a superb example of Chinese ingenuity, and it does employ an element that we might loosely call a pendulum, that is, the balanced column. It is, however, irrelevant to our discussion for several reasons. First, it could not display any of the dynamic qualities of pendular motion that became important in the West at a later time. Second, though it was in successful use in China for a long period of time, knowledge of its mechanical principles seems to have disappeared sometime after the sixth century; later writers knew of it only by reputation.[9] Third, there is no evidence that any scientific scholar, Chinese or Western, saw in the instrument any principles of dynamics that needed further elucidation. Finally, though the 'earthquake weathercock' might have been the prototype for crude, pendulum 'seismoscopes' in Japan in the nineteenth century,[10] there is nothing to indicate a spread westward,[11] or any further

[6] Needham *et al.*, *Science and civilisation*, vol. 3 (Cambridge: 1959), 626–635. (Note: for convenience I have adopted Needham's Romanization; other systems result in different spellings.) Sarton, *Introduction*, vol. 1 (Baltimore: 1927), 196 errs in referring this invention to Chhao Tsho and dating it 136 B.C.; compare Needham, vol. 3, 626, note 'e'.

[7] Akitune Imamura, 'Tyoko [that is, Chang Hêng] and his seismoscope', *Japanese journal of astronomy and geophysics*, **16** (1939), 37–42, first proposed the balanced column as a solution to some textual difficulties in the extant descriptions; a model reconstruction tested at the Seismological Observatory, Tokyo University, performed reasonably well. Imamura's success caused other students of the subject, especially Wang Chen-To, to abandon earlier reconstructions that had used a conventional, suspended pendulum. One still sees the earlier drawings of Wang in reproduced form. Needham, vol. 3, figs. 268–270 gives several versions.

[8] Owing to textual difficulties, the locking mechanisms proposed by Wang and Imamura differ in detail, but operate identically in principle. All reconstructions agree that the column must be locked in place after one motion only.

[9] Needham, vol. 3, 632–634. Hsintu Fang (fl. latter half sixth century) described the use of the instrument, while a contemporary, Lin Hsiao-Kung wrote a manual on its operation. A much later scholar, Chou Mi (fl. c. 1290) knew of their works only by name and expressed mystification over the way the seismometer might have worked.

[10] *Ibid.*, 635 refers to these as being 'of apparently traditional type'. This is difficult to evaluate, since only two Western-language reports seems to have appeared (Hugo Th. Horwitz, 'Beiträge zur Geschichte der aussereuropäischen Technik', *Beiträge zur Geschichte der Technik und Industrie*, **16** (1926), 294). Horwitz bases his information entirely on Gustav Kreitner, *Im fernen Osten: Reisen des Grafen Bela Szechenyi in Indien, Japan, China, Tibet und Birma* (Wien: Hölder, 1881), 293. These instruments consist of a pendulum with laterally projecting pins, all surrounded by a fixed paper cylinder; seismic tremors cause the pins to prick the paper. It would be most interesting to have a more reliable article on the use and ancestry of such a device.

[11] Needham, vol. 3, 634 notes that 'we hear of seismographic instruments at the Marāghah observatory'. No details as to the construction of these instruments are given, and I have been unable to find any descriptions in the literature. See André Godard, *Les monuments de Marāghah* (Publications de la Societe des Etudes Iraniennes et de l'Art Persan, No. 9; Paris: Leroux, 1934); and M. Fuad Köprülü, 'Maraga Rasathanesi hakkinda Bazi Notlar [some notes on the Marāghah Observatory: in Turkish]', *Belleten: revue publiée par la Société d'Histoire Turque*, **6**, nos. 23–24 (Jul.–Oct., 1942), 207–227 (*not* 'Bulletin of Ankara University' as in Needham, vol. 3).

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use of the device within the Western culture sphere. Thus, despite a promising beginning in the device of Chang Hêng, Chinese science and technology did not develop any further conceptions or instruments that have bearing on later Western thinking.

Turning to Islam, we find a persistent myth that the tenth-century astronomer, ibn Yūnūs, employed the pendulum as a time measuring device in his observations. Numerous popular histories of astronomy credit him, not only with the first use of the pendulum, but by implication with having some grasp of pendular isochrony.[12] Such claims turn out to have no foundation in fact and deserve to be laid to rest. Ibn Yūnūs's false reputation in this regard does not result from any claims he made, but from the misinterpretation of some diagrams in Arabic manuscripts at Oxford by Edward Bernard, astronomer and orientalist (d. 1696).[13] Bernard mentioned only briefly that the manuscripts showed the measurement of time *fili penduli vibrationibus*, but this remark was transmitted and amplified by generations of historians without, however, any reference to specific Arabic sources or texts.[14] What the great Cairene mentioned in his writings was a triangle apparatus with plumb bob to lay out a level surface on which astronomical instruments could be mounted. The proper interpretation was offered by the great German Islamicist, Eilhard Wiedemann, who added on the basis of a lifetime of familiarity with Muslim scientific writings that he knew of no text to mention the use of the pendulum before the modern period.[15] No one has contradicted this verdict in the past fifty years.

If both the pendulum *per se* and a theoretical concept of pendular motion are absent from classical antiquity, from the Chinese tradition and from Islam, is there any concept of oscillatory motion in general which could have had some influence on the late medieval scholastics? For the Aristotelian tradition, the

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Hugo J. Seemann, ' Die Instrumente der Sternwarte zu Marāghah nach der Mitteilungen von al ' ' Urdî ', *Sitzungsberichte der Physikalischen-medizinischen Sozietät zu Erlangen*, **60** (1928), 15–16 shows no seismographs, but does deal with several instruments using spirit and plumb-bob levels to align astronomical instruments. Since such leveling devices have been confused with the pendulum in another context (see text below), perhaps such is also the case in this instance.

[12] For example, Rudolf Wolf, *Geschichte der Astronomie* (München: R. Oldenbourg, 1877), 369; and P. Adolofo Müller, *Elementi di astronomia* (Roma: Desclée, 1904), vol. 1, 106. The fable lives on in the most diverse places, including P. R. Coleman-Norton, article ' Pendulum ', *Encyclopedia Americana* (New York: 1968), vol. 21, 502; and Nasr (footnote 5), 171.

[13] Edward Bernard, ' Letter of 28 March 1684 to Robert Huntington ', *Philosophical transactions of the Royal Society*, **14**, no. 158 (1684), 567. Strangely, Bernard mentions only one Islamic astronomer by name in this letter, Nasīr al-Dīn al-Tūsī, who was associated with Marāghah. I am still uncertain who injected ibn Yūnūs into the tale.

[14] The story appears in the writings of Laplace, Bailly and von Humboldt, to name but a few. Only Louis Sédillot seems to have expressed slight reservations in this regard.

[15] Eilhard Wiedemann, ' Ueber die angebliche Verwendung des Pendels zur Zeitmessung bei den Arabern ', *Verhandlungen der Deutschen Physikalischen Gesellschaft*, **21** (1919), 660–664; and his ' Ueber die angebliche Verwendung des Pendels bei den Arabern ', *Zeitschrift für Physik*, **10** (1922), 267–268. Wiedemann's expertise extended to the field of Islamic horology; see Eilhard Wiedemann and Fritz Hauser, ' Ueber die Uhr im Bereich der islamischen Kultur ', *Nova acta Academiae Caesar. Leopold-Carol. germ. naturae curiosum*, **100**, no. 5 (Halle: 1915), 112*ff*. Thus his denial of the ibn Yūnūs tale carries additional weight.

Since writing the above I have learned that Dr. David King, Director of the Project in Medieval Islamic Astronomy of the American Research Center in Egypt, has reached the same conclusion after a thorough review of all available evidence. See his article ' Ibn Yūnūs and the pendulum: a history of errors ', forthcoming in *Journal for the history of Arabic science*, **2** (1978).

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answer is largely negative.[16] For the other great ancient modes of thought, the Platonic and Stoic, the answer is more ambiguous, for there do seem to have been at least vaguely formulated notions that recur in several writers. Shortly before Socrates's death, as reported in Plato's *Phaedo*, he expatiates on the source of all terrestrial waters. The earth, he explains, is perforated by many subterranean passages, like the bowels of the human body, and these all connect with one enormous cavity at the earth's center, in which is found the mother of all waters, the lake Tartarus. Terrestrial rivers, lakes and seas all emerge from Tartarus and return to her in a vast cycle. Water is expressed from the earth's pores owing to the oscillation (αἰώρα) of Tartarus from one side of the globe to another, and since the center of the globe is hollow, there is nothing to stop Tartarus in her headlong perpetual onrush (ὁρμή).[17] Socrates does not explain why he thinks Tartarus should continue in motion about the Earth's center forever; he apparently assumes that, in the absence of anything to stop the motion, it will go on perpetually. Plato's passage could have served as the starting point for further speculative work on the nature of oscillatory motion, for it does contain, at least in embryonic form, the concept of a mass moving back and forth past a center of gravity or natural rest point, and it implies that such motion would have a natural period as well as a strong tendency to persist. Unfortunately, his vague manner of treating the subject led to destructive criticism from Aristotle, who with admirable common sense noted that water could not possibly flow uphill as well as downhill, that is, away from the earth's center as well as towards it.[18] Among less hostile writers, the *Phaedo* remark obtained a more sympathetic hearing, but it was

[16] Aristotle, *The physics* (ed. and trans. P. H. Wickstead and F. M. Cornford; 2 vols., The Loeb Classical Library; London: Heinemann, 1929–34), vol. 2, 370–373 (=VIII, 8, 262a, 13 *ff.*) discusses motion back and forth on a circular track, while pp. 392–395 (=VIII, 8, 264b, 24 *ff.*) considers the impossibility of moving continuously along a semi-circle or smaller arc, since this would require reversing one's motion. For a consideration of the sort of consistent doctrine of oscillatory motion one might reconstruct from these and similar general passages, see section 4 below. For our purposes, it is sufficient to note that Aristotle does not put forth such a doctrine nor make suggestions that would allow us to assume he had oscillating bodies or pendula in mind. The following passage from *De caelo* will illustrate the problem:

> all irregular motion has retardation, acceleration, and climax. The climax may be either at the source or at the goal or in the middle of the motion; thus we might say that for things moving naturally it is at the goal, for things moving contrary to nature it is at the source, and for things whose motion is that of a missile, it is in the middle.

On the heavens (trans. W. K. C. Guthrie; The Loeb Classical Library; Cambridge, Mass.: Harvard University Press, 1953), 170 (=288a, 19 *ff.*). Naturally, a motion quickest in its middle suggests to us a pendulum or similar oscillating body. If that was what Aristotle meant, he kept it a complete secret, even from the ancient commentators. We might echo Simplicius's comment here: '*τίνα οὖν ἐστιν τά ῥιπτούμενα*' ('What then are these "missiles"?'), quoted in Anneliese Maier, *Zwei Grundprobleme der scholastischen Naturphilosophie* (3rd ed.; Storia e letteratura, raccolta di studi e testi, vol. 37; Roma: 1968), 238.

[17] Plato, *Phaedo* (ed. John Burnet; Oxford: Clarendon Press, 1911), 111 e *ff.* Burnet was also of the opinion that Plato ascribed an oscillatory motion to the earth itself, holding that the globe moves up and down on its north–south axis. See his *Early Greek philosophy* (3rd ed.; London: Black, 1920), 303–304 discussing *Timaeus* 40 b 8 *ff.* The interpretation depends on Plato's meaning of the word *ἰλλομένην*. There has been some controversy over this passage, but the most accepted explication seems to be that of F. M. Cornford, that is that the earth 'winds round' its axis relative to the heavenly spheres (see his *Plato's cosmology: the Timaeus of Plato* (New York: Harcourt Brace, 1937), 120–134). No ancient commentator is known to have taken *Timaeus* 40 b 8 in Burnet's sense.

[18] Aristotle, *Meteorologica* (ed. and trans. H. D. P. Lee; The Loeb Classical Library; Cambridge, Mass.: Harvard University Press, 1952), 141 (=II, ii, 356a, 15–19).

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usually interpreted in hylozoic fashion on the analogy of inhalation and expiration as observed in animals.[19]

The only extant ancient writer who records a different use of Plato's notion in more concrete terms is Plutarch. Discussing the possible sphericity of the earth, he remarks at one point on how ridiculous it is to assume that 'incandescent masses of forty tons falling through the depths of the earth stop when they arrive at the center, though nothing encounter or support them; and if in their downward motion the onrush should carry them past the center, they swing back again and return of themselves'.[20] Equally laughable, Plutarch adds, is the idea that water flowing to the center of the earth 'stops suspended or moves about round it, oscillating in an incessant and perpetual see-saw'.[21]

Despite the fact that Plutarch includes such ideas merely to hold them up to ridicule, their appearance is proof that someone in antiquity was veering very close to a concept of oscillatory motion that included the germ of an intimation of inertia. Bodies carried past the earth's center by their onrush and returning to rest will reappear in Oresme's writings as part of a general idea that includes the pendulum. Unfortunately, we are unable to learn exactly what sources Plutarch drew upon for his 'ridiculous' ideas and images.[22] Either he created them himself or took them from now lost Stoic writings, which Plutarch could use freely because they were familiar to any reader of his dialog. If the latter is indeed the case, then we may assume further that Plato's inchoate conception of Tartarus's oscillations struck some sort of responsive chord among the Stoic physicists, who were often concerned with action at a distance and vibrational phenomena. Unfortunately, we must also assume that, however fertile this line of reasoning may have become in antiquity, it died out completely with the eclipse of Stoicism as a vital philosophy of nature toward the end of the classical period.

3. The 'discovery' of the pendulum

The foregoing series of negative judgments is meant to throw into sharp relief the real accomplishment of Jean Buridan and Nicole Oresme in bringing forth for the first time a view of the pendulum and of oscillating motion in general. Far from being a trivial exercise in the recognition of the obvious, their contribution to our story represents a genuine act of intellect that, in its own fashion, is as profound as Galileo's discovery of the law of isochrony two

[19] Olympiodorus described Plato's doctrine through a simile with the see-saw, ἀντιταλάντωσις, but he preferred an animistic explanation; compare Burnet ed. *Phaedo* (footnote 17), comments to 111 e 4. Cornford (footnote 17), 332 notes Olympiodorus's hylozoic explanation, as well as those of Athenodorus the Stoic and Pomponius Mela, and he suggests that an animistic analogy between the subterranean cavities with their aqueous flows and the bowels and lungs of animals may have been very close to Plato's own thinking here.

[20] Plutarch, *Moralia* (ed. and trans. H. Cherniss and W. C. Helmhold; The Loeb Classical Library; Cambridge, Mass.: Harvard University Press, 1957), vol. 12, 'De facie quae in orbis lunae apparet', 65 (=924 A–B). Chernis and Helmhold translate 'onrush', ῥυμῇ, as 'impetus'. I have emended the translation to the more literal meaning in order to avoid confusion with the scholastic doctrines discussed below.

[21] *Ibid.*, 924 B–C.

[22] The editors regard this and other strangely 'modern' notions in 'De facie' as merely a flight of Plutarch's rhetorical fancy. Sambursky, *Physical world of Greeks* (footnote 5), 209 *ff.* seems to regard them as a 'flash of intuitive genius' on Plutarch's part. Whether we follow either assumption, or neither, we still arrive at the same end point. Such ideas did not constitute part of an articulated theoretical structure of physical science that survived the collapse of the ancient world.

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and a half centuries later. The theoretical context in which the pendulum was 'discovered' is late medieval discussions of certain difficulties in the Aristotelian view of motion. For Aristotle, motion was a state of being that required some mover, not only to initiate it, but to sustain the motion at every moment in which it takes place. Inertia, as we understand it, played no role in Aristotle's physics; 'everything that is in motion must be moved by something'.[23] This 'something' could take a variety of forms; for example, the pull of an ox on an oxcart, or the inclination of a naturally heavy body such as a stone to revert to its 'natural place'. When the motive force or principle ceased to act on the moving body, the motion it had engendered also ceased. Within the realm of everyday experience, there was much to confirm such a common sense view of matters; when the ox ceased to pull, the cart stopped; when the falling rock hit the lowest possible point in its path, it also ceased to move.

Only one case seemed to contradict the general views Aristotle had put forth: that is, projectile motion, in which the moving body obviously continued in motion after its motive force had ceased. An arrow discharged from a bow was the most commonly used example. To meet these objections, Aristotle proposed two explanations, both more or less *ad hoc* and unsatisfactory in character. Either the air itself somehow transmitted a continuing force to the moving body, or the movement of air around the projectile somehow served to sustain its motion for a limited time. By the sixth century, an alternative theory had begun to take shape, albeit in primitive form. John Philoponus of Alexandria (fl. c. 520 A.D.) seems to have been the first to suggest that projectiles were moved by a force that the initial moving agent impressed into the moving object. This impressed force keeps the object in motion for a short time until it is overcome by the resistance of the body's weight (that is, the force is virtually self-exhausting), and, though Philoponus is unclear about this, the resistance of the medium in which the motion is occurring.[24] In both its self-exhausting and externally dissipated forms, Philoponus's *vis impressa* had some influence on several Islamic thinkers, including Avicenna and Alpetragius. Perhaps through these intermediaries, the idea was revived in Europe during the latter thirteenth century as Westerners for the first time began to explore fully the ramifications of the ancient philosophical writings to which they had fallen heir in the preceding century. We need not pause to explore the numerous arguments and counter-arguments, defenses and critiques that the *vis impressa* engendered among the scholastics[25] except to

[23] *Physics*, VII, 1, 241 b 24. It is possible to derive an incohate concept of inertia from Aristotle by pressing very hard on the paradoxical consequences of his argument against the existence of a void. See Edward Grant, 'Motion in the void and the principle of inertia in the Middle Ages', *Isis*, **55** (1964), 265–292.

[24] Ioannes Philoponus, *In Aristotelis physicorum libros commentaria* (ed. H. Vitelli), in *Commentaria in Aristotelem graeca*, vol. 17 (Berlin: 1888), 639.3–642.9. A translation of the passages is included in M. R. Cohen and I. E. Drabkin, *A source book in Greek science* (Cambridge, Mass.: Harvard University Press, 1948), 221–223. See also Walter Böhm, *Johannes Philoponus: christliche Naturwissenschaft im Ausklang der Antike: ausgewählte Schriften* (München: Schöningh, 1967).

[25] The literature on this subject is impressive in its bulk and complexity. I have found the following major works especially useful. Pierre Duhem, *Etudes sur Léonard de Vinci* (2nd series; Paris: Hermann, 1909), and (3rd series; Paris: Hermann, 1913) should be carefully compared against his massive ten-volume *Le système du monde* (Paris: Hermann, 1914–59). The many works of Anneliese Maier constitute a careful reworking of Duhem and an alteration of some of

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note that, in slightly modified form, it was strongly championed in the middle decades of the fourteenth century at the University of Paris.

The Parisians' version of the *vis impressa*, known as *impetus*, was not in their minds a hostile or destructive means of attacking the fundamentals of Aristotelian physics. It was rather an adjustment of existing theory to create a better fit with certain cases that were awkward or impossible to treat under classical Aristotelian canons. It was in the course of extending theory into previously unconsidered cases (and thereby bolstering the argument in favor of *impetus*) that the pendulum makes its first appearance. Nicole Oresme (d. 1382) was active at the University of Paris in the 1350s when he composed a brief commentary in Latin on Aristotle's *De caelo*; considering how *impetus* can be employed to explain certain observed motions, he notes:

> Further, this [*impetus*] can explain other experience regarding this motion. One is that a heavy thing is moved farther than its own natural place, as is evident in the case of a [ball of] lead hanging from a cord, and it would be true in the case of a stone descending to the center of the world, if the latter were wholly perforated, as [the stone] would pass beyond [the center] and after many motions back and forth, would rest. The cause of this can be assigned to this *impetus* because at that time it is stronger than the resistance.[26]

In later life, as bishop of Lisieux, Oresme was commanded by King Charles V to do a French translation of and commentary on *De caelo et mundo* for the edification of his courtiers who did not read Latin. Oresme obliged his monarch in 1377, and in the course of this work he returned to the idea he had enunciated briefly twenty years before. Considering the 'weight' that a falling body acquires (the modern 'momentum'), he objects that

> this property or quality may be called impetuosity, and it is not properly called weight or heaviness. For if an opening were made from here to the center of the earth and beyond and a heavy object fell through this opening or hole, upon reaching the center it would pass beyond and begin to go upward by reason of this accidental and acquired property; then it would fall back again and come and go several times just as we can observe in the case of a heavy object hanging from a beam by a long cord. Therefore since this property causes a heavy body to move upward, it definitely is not the same as weight or heaviness.[27]

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his less well-considered conclusions. In chronological order these are *Die Impetustheorie der Scholastik* (Wien: 1940), *Die Vorläufer Galileis im* 14. *Jahrhundert* (Roma: 1949), *An der Grenze von Scholastik und Naturwissenschaften* (2nd ed.; Roma: 1952), and most importantly, the 3rd edition of her *Zwei Grundprobleme der scholastischen Naturphilosophie* cited in footnote 16 above. Briefer discussions are found in Marshall Clagett, *The science of mechanics in the Middle Ages* (Madison: University of Wisconsin Press, 1959); and A. C. Crombie, *Augustine to Galileo* (2nd ed., 1959); I have used the 2 vol. ed. (Harmondsworth: Penguin, 1969). Ernest A. Moody, 'Galileo and Avempace', *Journal of the history of ideas*, **12** (1951), 163–193 and 375–422 is also useful.

[26] Claudia Kren, 'The *Questiones super de celo* of Nicole Oresme' (unpublished Ph.D. Dissertation: University of Wisconsin; Madison: 1965), 363–366 (=Book II, question 7, lines 393 *ff.*): the English translation is Kren's. An abbreviated version of the same is found in Maier, (footnote 16), 247. For the probable dates, compare Kren, xi and Maier, 385. Question 7 concerns whether 'natural motion is faster in the end than in the beginning'; it concludes that it is and that *impetus* is the explanation.

[27] Nicole Oresme, *Le livre du ciel et du monde* (ed. and trans. Albert D. Menut and Alexander J. Denomy; Madison: University of Wisconsin Press, 1968), 144–145 (=Book I, chapter 18). The English translation is Menut's. Compare also Clagett (footnote 25), 570–571; Maier (footnote 16), 252; and Dulem *Etudes* (footnote 25), vol. 3, 351–353 and *System*, vol. 8, 300–301.

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Finally, later in the same work Oresme puts forth his most extended discussion of the topic:

> I posit that the earth is pierced clear through and that we can see through a great hole farther and farther right up to the other end where the antipodes would be if the whole earth were inhabited; I say, first of all, that if we dropped a stone through the hole, it would pass beyond the center of the earth, going straight on toward the other side for a certain limited distance and that then it would turn back going beyond the center on this side of the earth; afterward it would fall back again, going beyond the center, but not so far as before; it would come and go this way several times, but with a reduction of its reflex motions until finally it would rest at the center of the earth. This is caused by the impetuosity or ' momentum ' which it acquires by the acceleration of its motion. . . . We can understand this more easily by taking note of something perceptible to the senses: if a heavy object *b* is hung on a long string and pushed forward, it begins to move backward and then forward, making several swings, until it finally rests absolutely perpendicular and as near the center as possible.[28]

In the original, this text is accompanied by a marginal drawing, the first known representation of a pendulum.

We may begin to analyse Oresme's remarks by looking at the work of his most famous predecessor, Jean Buridan (d. 1358), who almost certainly was Oresme's teacher at the University of Paris.[29] Although Buridan's extant works contain no explicit discussion of the pendulum or of the ' hole through the earth ' problem, they do abound with references to *impetus* and to examples of motion in which *impetus* plays a necessary explanatory role. Although Oresme certainly was the more elegant stylist of the two, Buridan's work is marked by a tendency toward concrete, homely images from the world of everyday experience. In his commentary on the *Physics*, for example, he argues against Aristotle's air-push theory of continuing motion by citing such cases as a spinning top and a rotating grindstone, where motion patently persists without the involvement of the surrounding medium. In his further remarks he observes:

> Also, it is this way with a cither cord which, put under strong tension and percussion, remains a long time in a certain vibration from which its sound continues a notable time. And this takes place as follows: as a result of striking [the cord] swiftly, it is bent violently in one direction, and so it returns swiftly toward its normal straight position. But on account of the *impetus*, it crosses beyond the normal straight position in the contrary direction and then again returns. It does this many times. For a similar reason a bell, after the ringer ceases to draw [the cord], is moved a long time, first in one direction, now in another. And it cannot be easily and quickly brought to rest.[30]

[28] Oresme, *Du ciel*, 572–573 (=Book II, chapter 31).

[29] Buridan was at the University from at least 1328 until after 1356. Oresme was a student there in 1348 and a master by 1356 at the latest (see Clagett (footnote 25), 337 *ff.* and 521 *ff.*). Kren (footnote 26), vi notes that Oresme never mentions Buridan in his extant writings, but Buridan refers to Oresme in his *Quaestiones* on Aristotle's *Meteorology* (*post* 1356).

[30] Trans. Clagett (footnote 25), 537–538. Compare Maier (footnote 16), 214 (=Book VIII, quest. 12). There is no modern edition of the original. Interestingly, Buridan's earlier commentary on *De caelo* contains many discussions of *impetus*, but no mention of the problem of oscillatory motion, despite the editor's mistaken assertion that it does; see Ernest A. Moody (ed.), *Johannis Buridani quaestiones super libris quattuor de caelo et mundo* (Cambridge, Mass.: Mediaeval Academy of America, 1942), xx. Apparently, by about 1340, Buridan still had not extended *impetus* into the realm of oscillatory movement. Crombie (footnote 25), vol. 2, 86, apparently following Moody, errs in claiming Buridan refers to the ' oscillation of a pendulum '.

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While Buridan's imagery is concrete and Oresme's abstract, the two are referring to very similar physical phenomena. The vibrating either string, like the weight falling through the earth, oscillates on either side of an equilibrium or rest point until its motion is damped out.[31] The parallel between Buridan's bell and Oresme's pendulum is obvious; the latter is merely an abstracted version of the former, reduced to its essential components, a weight suspended from a cord. (We should note here that Buridan's reference means the entire bell, housing and yoke included, and not merely the suspended clapper.) We can date with reasonable precision the transformation of bell to pendulum, since one of Oresme's earlier writings also contains the 'hole in the earth' problem. In his *Quaestiones de sphaera*, done shortly before the Latin *De caelo* commentary, the matter is treated as a separate question, *Utrum grave particule sicut est lapis is non esset impeditum, moveretur ad centrum, et ibi quiesceret naturaliter*. The answer is, of course, yes, but in explicating the question Oresme seeks a visible analogy to the stone. It acts, he says, 'exactly as when a javelin, if it is dropped to the earth, still quivers, although it would be in a place where it could be at rest. And just as in a balance, when one arm is raised and released, it oscillates around the center many times'.[32] Thus at the time *De sphaera* was composed Oresme had already begun to absorb impetus theory and some of its basic implications, but apparently he had not assimilated Buridan's bell example into his thinking. He was forced to select less satisfactory examples to serve as visible counterparts to the stone's motion through the earth. By the early 1350s, when he wrote the *De caelo* questions, he was markedly more familiar with Buridan's doctrines, including the latter's exemplary bell.[33]

[31] Of course, the actual behavior of a vibrating string is much more complex than Buridan could have realised. It is described by the equation:

$$m\frac{d^2\xi}{dt^2}+R\frac{\partial\xi}{\partial t}+f\xi=0,$$

where m = inertial mass, R = damping factor, f = elasticity of system (tendency to return to equilibrium), and ξ = displacement from equilibrium. Vibration cannot take place unless

$$f>\frac{R^2}{4m}.$$

Even intuitively, Buridan's explanation seems too limited, since it cannot explain why some strings vibrate more rapidly than others. Perhaps some intuitive considerations of this sort influenced Oresme to substitute the simpler 'hole in the earth' image to exemplify the same point.

[32] Garrett Droppers, 'The *Questiones de spera* of Nicole Oresme' (unpublished Ph.D. Dissertation, University of Wisconsin; Madison: 1966), 82–97 (=Question 4). The English translation is Dropper's. On the date see *ibid.*, 6–7; and Maier (footnote 10), 386.

[33] In *De sphaera*, Oresme does not explain the falling stone's motion exclusively by means of *impetus*, but by presenting an Aristotelian 'air push' theory and Buridan's *impetus* without choosing between them (Droppers *ibid.*, 354 *ff.* commenting on text in footnote 32 above). This conforms to Maier's contention that Oresme's earliest works show only faint traces of Buridan's influences, whereas later Oresme accepted his teacher's doctrines more fully, though in modified form. If we accept Maier's dating of the written version of Buridan's *Physics* commentary at 1355–1357 (see her (footnote 16), 207), then Oresme's earliest discussion of the pendulum is slightly too early to have been influenced by Buridan's passage. However, since both men were at the University of Paris simultaneously, Oresme could have heard of Buridan's ideas first in lecture form. Also, since the dating of Oresme's Latin *De caelo* commentary is very uncertain, it could be slightly later than Buridan's *Physics* commentary. Another early Oresme work, his commentary on Aristotle's *Physics*, was found in an incomplete manuscript in Spain in 1964. Maier, *ibid.*, 382–385 describes its contents. Apparently the 'hole in the earth' is included, but not the pendulum. Thus by every indication available, both Buridan and Oresme moved toward applying *impetus* to oscillating motion systems and thus 'discovering' the pendulum only in the 1350s.

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4. The discovery and the philosophers

Further analysis of the accomplishment of Buridan and Oresme could take two paths: historical tracing of the roots from which their ideas grew, or more philosophical consideration of the problem in 'historical epistemology' that their discovery presents. Since the latter has been the mode of some discussion, we shall consider it first.

We would seem to have in hand an example of a type of problem the historian encounters from time to time, a matter of the relationship between what we earlier called 'knowing' and 'seeing'. If we assume that the pendulum was always in reality a simple entity, then it is reasonable to assume that it should have been recognized (in some sense of that word) long before it was. It seems that the pendulum was always 'there' to be perceived, yet we know that a large group of thinkers (presumably without a marked proclivity toward myopia) did not 'see' it. At a certain point in time, certain thinkers 'discovered' what was always 'there'. The difference between those who did not 'see' the pendulum and those who did would seem necessarily to lie in the state of their 'knowing'. In other words, the failure to 'see' the pendulum must in turn be related to some difficulty in integrating it into a pre-existing matrix of concepts or 'mental set'. As long as certain beliefs about motion were generally held, the pendulum was invisible; when these had changed in some (unspecified) fashion, the pendulum became visible. Reducing the entire matter to a formula: 'knowing' precedes and predetermines 'seeing'.

There is ample evidence to support the formula. Several generations of perceptual psychologists have shown time and time again how our minds project complete images on to incomplete patterns, how we 'see' what is not 'there'. Conversely, an impressive body of experimental evidence shows how our minds exclude information that we think conflicts with its surrounding pattern or the entire *Gestalt*, how we refuse to 'see' what is 'there'. Perception is by no means a passive state, but rather an active, constant transaction between the observer, his environment, and the observed entity. Surely it would be a naive historian indeed who could assert that all men ever had to do was to see what was before their eyes. Perhaps the most famous example from the history of science to confirm the priority of 'knowing' over 'seeing' concerns the stars themselves. From time to time new stars make an appearance in the heavens, usually remaining visible for several months or even years before fading from view. Such novae (or supernovae in the most brilliant cases) were rarely observed by Europeans during the nearly two millenia that Aristotle's dictum of celestial immutability was unchallenged. Chinese astronomers, however, blissfully unaware of how important it was that the heavens be unchanging, observed numerous novae during this same period (indeed, their observations are the basis for our knowing that such phenomena did occur). In the West, only after Copernicus's doctrines had begun to undermine confidence in the Aristotelian cosmos did novae appear to the eyes of Europeans in anything like an impressive fashion.[34]

[34] Kuhn (footnote 2), 116. The Chinese and Western treatments of sunspots provide an almost exactly parallel case.

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The treatments of the scholastic pendulum by Thomas Kuhn and Piero Ariotti both rest on the belief that 'knowing' holds priority over 'seeing'. For Kuhn, the pendulum's initial appearance is yet another example of the power that 'paradigms' (read 'mental sets' or 'conceptual matrices') have to control what a scientist 'sees'. The pendulum, he argues, was far from simple if one tried to analyse its motion within the framework of ideas concerning motion held by a conventional Aristotelian before *impetus* theory appeared. So great was the complexity of such an analysis, he contends (more in relation to Galileo than the scholastics), that any attempt to come to grips with the pendulum would have been tantamount to direct confrontation of a serious anomaly, and would have carried with it the threat of provoking a crisis within the entire structure of Aristotelian physics. Hence the pendulum was shunned until a modified Aristotelianism had created in *impetus* a theoretical construct capable of treating the pendulum in safe ways. Thereupon the pendulum simply appeared from obscurity. 'Until that scholastic paradigm was invented', he writes, 'there were no pendulums, but only swinging stones, for the scientist to see. Pendulums were brought into being by something very like a paradigm-induced gestalt switch'.[35] Quite literally on this view the pendulum did not exist and could not exist as an object of 'seeing' until it could be somehow 'known'.

Kuhn's explanations of the scholastic pendulum are but a small part of his much-discussed theories about the nature of scientific change; thus he may be excused for being slightly vague on this specific case. No such difficulty arises in the work of Piero Ariotti, a philosopher who seems not to share Kuhn's views in every respect, but who still falls within the general pattern outlined above. For Ariotti, the salient feature of pendular motion is its continuity. Peripatetic physics, he argues, could not possibly have grasped continuous oscillating motion because of its pervasive habit of analysing motion into pairs of logical contraries; for example, back and forth, up and down, and so on. Moreover, Aristotelians had no notion of instantaneous changes of state because of the way they construed what he calls the 'principle of the excluded middle'. Hence an Aristotelian faced with a pendulum would instinctively analyse its motion into upward and a downward component, and he would insist that there were three distinct states of rest in each demi-oscillation—at each end of the swing and at the midpoint, where downward movement becomes upward movement. *Impetus* theories did not, of course, abandon these distinctions, but Ariotti contends that they did blur them to such a degree that the pendulum could for the first time be treated as a soluble problem.[36]

Both views may be subjected to similar criticisms. Neither seems entirely wrong, but both are inadequate as explanations for the pendulum's discovery. Both offer some explanation for the failure of Aristotelian physics to consider the pendulum. Kuhn's is couched in very general terms: the pendulum was a 'difficult' case. Ariotti boldly reconstructs a hypothetical Aristotelian position on the subject; it is a good argument, but it begs the question, since

[35] *Ibid.*, 120.
[36] Ariotti, 'Development of pendulum' (footnote 2), 333–340.

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no Aristotelian ever advanced it at all.[37] More important is the fact that both men deal only with the Aristotelian tradition, whereas the pendulum lay equally unrecognized by Platonists, Stoics, Chinese thinkers, and everyone else. Any explanation for the pendulum's non-recognition must be broad enough to encompass more than just Peripatetic thought. Equally, both Kuhn and Ariotti leave us with little help in understanding how Buridan and Oresme were able to make the transition from the general to the specific, from *impetus* theory to the pendulum as a concrete example of that theory. Kuhn takes refuge in 'something very like a paradigm-induced gestalt switch'. That is, he substitutes a psychological mystery for an historical problem. Ariotti's explanation is equally weak in asserting that *impetus* theory blurred the old distinctions between logically contrary categories of motion. Perhaps it did, but a reading of Oresme's remarks indicates that the opposite is the case. That is, the distinctions are as sharp as they ever were, but the pendulum provides an example in which they seem to be violated; on dynamic rather than kinematic grounds, this violation is explicable in terms of *impetus*. Kuhn and Ariotti have answered the question of the pendulum's discovery in ways that raise still deeper and less soluble problems than the original one. In part, this results from their misconstruing the epistemological aspects of the problem through a failure to consider the relevant historical information. We shall return to the problem of 'knowing' and 'seeing' shortly, after having glanced at the historical roots of Buridan and Oresme's thoughts.

5. The background: theory, image, model

It is possible to view the pendulum's emergence as the result of a convergence of ideas, images and physical things within the framework provided by *impetus* theories. That theory provided any interested thinker with the means to treat old problems in new ways—indeed, that is certainly why it developed in the first instance—but also to treat new problems that had been uncommon, if not wholly ignored, in the old framework. Chief among these was a class of problems involving persistence of motion other than just the classical problem of projectile motion. In Buridan's writings, for example, we find him considering such instances as the persistent surge of a boat after its oarsmen have ceased to pull, the spinning of a top and a grindstone, and the bouncing of a tennis ball, as well as the above mentioned cither and bell. Indeed, there is an implied progression in Buridan's examples leading from the commonly invoked instance, simple projectile motion, to the more elaborate cases, especially reflexive and oscillatory motion. The last case is especially significant, because, as Oresme observes, it involves motion that persists into a state where it should not, that is, upward motion resulting from an initial downward movement. The point was not to 'blur the distinctions' between the conventional categories of motion (which Buridan and Oresme seem to accept as thoroughly as any philosopher of their day), but to illustrate how

[37] Some discussions of the problems of discontinuities in 'reflexive' motion do exist (see Duhem, *Etudes* (footnote 25), vol. 3, 127–128 and 144–145). Droppers (footnote 32), 101 *ff.* and 362 *ff.*, deals with Oresme's discussion of the matter. Leonardo da Vinci's observations of the motion of the pendulum are most succinctly given in *Codex Madrid* I, fol. 147r (see Ladislao Reti (ed. and trans.), *The Madrid codices* (New York: McGraw-Hill, 1974); and (footnote 2), *passim*). None of these examples, however, contains the sort of logical obstacles that Ariotti emphasizes; indeed, they all stem from the period after *impetus* theories had become widely known.

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impetus must be seen as a distinct cause of motion in its own right, quite apart from mere heaviness or directly forced movement. In other words, the goal seems to have been to sharpen the definition of *impetus*, to assure its autonomous status within a modified Aristotelian framework, and to demonstrate its applicability to a wide range of phenomena.

How then are we to account for the appearance of the pendulum as a specific case worthy of consideration within this general attempt? The answer lies in a consideration of the sources from which Buridan and Oresme drew the images they used in presenting their doctrines, namely, the ' hole in the earth ' image and that of the swinging bell. The former seems to have developed out of a purely speculative means of illustrating a completely orthodox Aristotelian tenet, the existence of a spherical earth in the very middle of a spherical universe and the fact that it rests there without any support whatsoever. For Aristotle, this is the inevitable logical consequence of his doctrine of natural place. Each of the four elements has its own proper degree of lightness or heaviness, and left to themselves, each would seek its own special place in the spherical cosmos. The earth element, as the heaviest of the four, is the most densely compacted; it occupies the solid of least volume, a sphere, and seeks the maximum distance possible from the lightest element, fire. Since fire tends toward the outer periphery of the spherical cosmos, the earth must occupy the center of the universe, the one spot equidistant from all points on the cosmic sphere's surface. Thus, any particle of the earth element, if fully unconstrained, would naturally tend toward the center of the universe, there to remain in perpetuity.

The only image with which Aristotle himself invests this idea occurs in *De caelo*, where he (for heuristic purposes only) imagines the earth in the process of formation out of the primordial chaos in which all the elements would be homogeneously distributed throughout the cosmic globe. The earth element would naturally converge on the center of the universe to form a sphere whose center was concentric with the center of the universe, just as individual particles of earth now tend to fall toward that same center.[38] (Aristotle is aware that falling bodies move in slightly convergent, not parallel lines.) A more dramatic means of illustrating the same principle makes its first appearance in a dialog composed in the early twelfth century by the English philosopher, Adelard of Bath, *Quaestiones naturales*. The work takes the form of a freely moving conversation between Adelard, an ebullient, well-traveled man, and his inquisitive nephew. Cosmology enters the talk, and Adelard explains in proper Aristotelian fashion how it is that a spherical earth occupies the center of a spherical universe without need of any prop to keep it there. The nephew then asks (freely translated):

> Supposing the earth to be pierced so that there were a passage from one heaven to the other, and a stone be dropped in; what would be the direction of its fall?
>
> *Adelard:* That which causes the stationary position of the earth would bring the stone to rest.

[38] *De caelo* (Guthrie trans., 243–251 (=II, xiv, 296 b 6–297 b 24)). The same general explanation appears in compressed form in the most popular of medieval astronomical works, John of Sacrobosco's *Tractatus de sphaera*. See Lynn Thorndike, *The sphere of Sacrobosco and its commentators* (Chicago University of Chicago Press, 1949), 84–85 and 122.

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> *Nephew:* The answer satisfies me, for I understand that it would come to rest at the central point.[39]

Adelard goes on to explain that this is due to the tendency of each of the elements to flee from their contraries to the greatest degree possible, causing the stone to seek a place as far away from the firey heavens as it could possibly occupy.

Adelard created in this dialog (and in a similar work, *De codem et diverso*) the core of the image that Oresme employed in illustrating the effects of impetus. We have no means of knowing whether Oresme was directly indebted to the Englishman or not, but we can hear what appear to be echoes of Adelard's question in other scholastic writings. What would happen if a heavy (earthly) body found itself moving toward the center of the earth (the center of the world)? Since at least part of such a body would have to pass beyond the earth's center in order for its own center to coincide with the earth's center, would the motion of the part that had passed beyond the center be ' natural ' or ' unnatural '?

Walter of Burley, a Fellow of Merton College, answered early in the fourteenth century that such a motion would be indeed ' natural ' in reference to the body as a whole, though it might appear unnatural for the body considered only in reference to its parts. A later Mertonian, Richard Swineshead, took up the question in a somewhat different manner in Tractatus XI of his *Liber calculationum* (written probably between c. 1340 and 1350). Would the part of the body that had passed the center offer resistance to the part that had not yet passed the center, and if so to what degree and with what consequences? Following the laws governing proportionality stated by Thomas Bradwardine and with extreme mathematical ingenuity, Richard seeks to compare the proportion of motive force to that of resistance as a function of the ratios of the lengths of the body on either side of the center point; he does so in both step-function fashion and (with even greater difficulty)

[39] Martin Müller, ' Die *Quaestiones naturales* des Adelardus von Bath ', *Beiträge zur Geschichte der Philosophie und Theologie des Mittelalters*, **31**, Hft. 2 (München: 1934), 48–49 (=Quest. 49). The translation is, with modifications, that of Hermann Gollancz (ed. and trans.), Berechiah ben Natronai, ha-Nakdan, *Dodi ve-Nechdi* [Uncle and nephew: a medieval Hebrew version of the *Quaestiones naturales*] (London: Oxford, 1920), 138. Gollancz translates both the Latin and Hebrew versions for comparative purposes. Is Adelard's image some pale ghost of Socrates's Tartarus? Perhaps, but the source of his conception is not known. Müller's suggestion (p. 86) that it is derived from Chalcidius's Latin *Timaeus* is untenable, and *Phaedo* was not translated into Latin until 1156 (Sarton (footnote 5), vol. 1, 113). In a related work, *De eodem et diverso*, Adelard places the nephew's question in the mouth of a Greek philosopher from Salerno, without, however, indicating an answer (see H. Willems, ' Der Adelard von Bath Traktat *De eodem et diverso*, zum ersten Mal herausgegeben und hist.-krit. untersucht', *Beiträge zur Geschichte der Philosophie des Mittelalters*, **4**, Hft. 1 (Münster: 1903), 33). This may be Adelard's way of indicating that he learned of it during his long travels in Spain and Italy, perhaps from Greek or even Muslim sources. A statement expressing Adelard's thought appears in the work of Al-Khāzinī on statics: 'A heavy body moving toward the center of the world does not deviate from the center; and when it reaches that point its motion ceases ' (see Nicolai V. K. Khanikoff, 'Analysis and extracts of . . . the *Book of the balance of wisdom*, an Arabic work on the water-balance, written by al-Khāzinī in the twelfth century ', *Journal of the American Oriental Society*, **6** (1860), 30). Al-Khāzinī presents this as a fundamental theorem in a list of such theorems that he attributes to Abū-Sahl al-Kūhi (fl. about 988) and ibn al-Haitham (Alhazen; c. 965–c. 1039). Thus it would seem to have been fairly widespread among Muslim natural philosophers.

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in continuous-function fashion. Both results lead him to the same paradoxical conclusion: in no finite time can the center of the body and the center of the world coincide. This is absurd for Richard, however, so he argues further in the manner of Burley that the body in question cannot be treated as merely the sum of its parts, but if one considers the *entire* body its center will obviously come to rest at the center of the world. Indeed, Richard's principal point in making the argument seems to be the distinction between the body viewed as the sum of its parts and the body viewed as more than the sum of its parts.[40] Moreover, neither Richard nor any of the other English discussants of this problem seems to have considered the possibility that the body might oscillate about the center. For this reason, we cannot see Swineshead as Oresme's immediate precursor in respect to the pendulum problem. To be sure, when one compares the simplicity of Oresme's treatment with the complexity of Richard's, it is tempting to claim that Oresme sought to use *impetus* as a way around the paradoxical conclusions of the Mertonians. In the present state of the evidence, that would be pure speculation. All we can securely assert is that Adelard's ' hole in the earth ' image had become an imaginary means to frame and visualize certain problems for discussion by the fourteenth century,[41] that Oresme was generally familiar with this tradition of thought, and that he adopted the image for his own argumentative purposes.

The other important element in the appearance of the pendulum was a device that would display the persistence of oscillatory motion in a clear and dramatic fashion, the heavy suspended bell. Buridan's remark can only refer to one sort of bell, the type still common today which is suspended from a pivoted yoke and rung by being moved back and forth. No Greek or Roman ever saw such a machine. Almost all ancient bells are small (*tintinabula* in the Latin diminuitive), measuring less than ten cm. in height and light enough to be carried about by hand. Larger bells are characteristic only of Latin Christendom in the middle ages and China. The early history of bells in both

[40] M. A. Hoskin and A. G. Molland, ' Swineshead on falling bodies: an example of fourteenth-century physics ', *British journal for the history of science*, **3** (1966), 150–182. Compare A. G. Molland, ' The geometrical background to the " Merton School " ', *British journal for the history of science*, **4** (1968), 121–123; also J. E. Murdoch and E. D. Sylla, ' Swineshead, Richard ', in C. C. Gillispie (ed.), *Dictionary of scientific biography*, vol. 13 (New York: Scribner's, 1976), 198 *ff*. Lynn Thorndike, *A history of magic and experimental science*, vol. 3 (New York: Columbia, 1934), 381 *ff*. also mentions the problem. Swineshead reaches his paradoxical conclusion by dividing the distance between the center of the earth and the center of the body into an infinite number of parts, each half of its predecessor; each part takes longer to traverse the center than its preceding part. Thus there is a regression to infinity without any resolution. The continuous-function method is more difficult to describe, particularly since Richard must make do without the calculus, but it yields the same result (see Molland, ' Geometrical background ' *supra*, 122). It is possible that Richard was far more interested in these modes of treating the problem than in the final answer, which has the ironic quality of denying that the problem is tractable by mathematical means.

[41] Buridan's *De caelo* commentary includes the image of a hole through the earth as a means of illustrating the natural abhorrence of a vacuum. Not only would an earthen substance naturally fall through the hole, but so would water, or air, or even fire rather than leave the hole a complete void. *Impetus* does not enter into this discussion at all (Moody ed., 267) (=Book IV, quest. 7)). Oresme made the matter of a stone's falling through such a hole a separate question in his *De sphaera* commentary; compare footnotes 32 and 33 above. Clearly the image existed apart from the specific questions surrounding persistence of motion and *impetus*.

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cultures is still a matter of shadows and guesswork,[42] but in both there is a general trend toward larger and larger devices. A large bell has acoustical properties that differ from those of a small bell, most particularly in the tendency of large bells to fracture from internal vibrational stress if improperly rung (for example, the American 'Liberty Bell'). The Chinese approached this problem by mounting large bells, even huge examples, on a fixed support and striking them from the outside with a separate clapper.[43] Western bells, with their internal clappers, evolved a different pattern, called 'live hanging' by modern bellfounders. The bell is mounted on a yoke, the yoke is suspended from two bearings, and the entire mass is swung back and forth to allow the clapper to strike the inner lip of the housing. We are not certain just when this system became commonplace in Europe;[44] the first unambiguous reference to it comes from Theophilus Presbyter's *De diversis artibus* of 1122, where it is fully described without any indication that it was a technical novelty.[45] Live hanging alone would not produce the motion that Buridan mentions; for that an exceptionally heavy bell is needed. We can trace a steady increase in the maximum possible size of bells in western Europe from about the year 1000, when about 60 cm. and 130 kg. seem to have been the upper limit, to the fifteenth century, when monster specimens of over 11,000 kg. were cast.[46] As bells increase in weight, their inertial mass becomes disproportionately greater than the friction exerted by their bearings, and thus they have a greater tendency to continue swinging after the bell-ringer ceases to pull on the rope. Such a bell offers a truly impressive example of energy storage. It can be set in motion by reasonably light pulls on its rope if they are delivered in a properly timed series; once in motion, however, the bell is impossible to stop quickly. Should an unwary sexton grasp the rope at the wrong time, he will be lifted off his feet into the air. These are the sorts of phenomena

[42] Surveys of the early history of bells are found in Feldhaus (footnote 3), cols. 462 *ff.*; Karl Walter, *Glockenkunde* (Regensburg: Pustet, 1913), 17–41; and Heinrich Otte, *Glockenkunde* (2nd ed.; Leipzig: 1884), *passim*. The articles by H. Leclercq, 'Cloche', and L. Gougaud, 'Clochettes celtiques', in *Dictionnaire d'archéologie crétienne et de liturgie*, vol. 3, part 2 (1914), cols. 1977 *ff.* and 1982 *ff.* respectively are still useful. Joseph Smits van Waesberghe, *Cymbala: Bells in the Middle Ages* (Roma: 1951), though dealing mainly with glockenspiels, does give some additional information. Needham, *Science*, vol. 4, part 1 (Cambridge: 1962), 195 *ff.* gives some information on Chinese bells.

[43] Apparently there are two types of Chinese bells: *Chung*, which are large types with external clappers; and *Ling*, which are smaller (up to about 40 cm.) with internal clappers (see A. C. Moule, 'Chinese musical instruments', *Journal of the North China Branch of the Royal Asiatic Society*, **29** (1908), 35–46).

[44] All the early medieval references that might indicate a 'live hanging' fail to specify whether the movement mentioned is of the entire bell or merely the clapper. Otte (footnote 42), 9–14, discusses several pieces of evidence, and reaches the conclusion that large bells, regardless of suspension type, did not themselves become common in Europe before the ninth century, implying that widespread use of 'live hanging' was even later. E. Viollet-le-Duc, *Dictionnaire raisonné de l'architecture française du XIe au XVIe siecle*, vol. 3 (Paris: Bance, n.d. [about 1865]), 286, notes that the towers of most early churches were too narrow to allow a bell to swing, and the belfrys (framework within the tower) seem not to have been constructed to permit live hanging. No one has, to my knowledge, determined a specific date for 'live hanging', but it does not seem to have appeared early in the middle ages.

[45] Ed. and trans. C R. Dodwell (London: Nelson, 1961), 157–158; compare trans. J. G. Hawthorne and C. S. Smith (Chicago: University of Chicago Press, 1963), 175–176. On the date, see Lynn White, jr., 'Theophilus redivivus', *Technology and culture*, **5** (1964), 230.

[46] Otte (footnote 42), 138. The great bell of Cologne, for example, cast in 1437–1438, weighed 250 hundredweight, or roughly 12,700 kg.

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Buridan must have had in mind when choosing the bell as an example, commonplace occurrences that would have been part of the everyday experience of anyone at an urban university, but also splendidly dramatic examples of *impetus* in action.

6. Analysis and explanation

With these elements of historical background in mind, we can construct a new explanation for the pendulum's emergence into philosophical consciousness in the fourteenth century. Previous analyses have left a certain air of mystery to the subject in large measure because they tacitly assumed that something as ' simple ' as the pendulum must have been ' seen ' at an early stage in human history. Since it manifestly was not, then one has the task of explaining how the obvious lay unrecognized; changes in theory or the state of ' knowing ' are held to have enabled the envisioning of the pendulum as a significant case of repetitive motion. An alternative approach emphasizes the pendulum's appearance as the result of a combination of theoretical changes, the existence of a striking, but partly puzzling image (the ' hole through the earth ' thought-experiment), and what we must call the proximate physical model (the heavy suspended bell). On this view, the pendulum became visible in two stages, the first represented by Buridan, the second by Oresme. *Impetus* was, of course, the indispensable theoretical construct out of which the pendulum emerged, for it alone was capable of offering a satisfactory explanation of persistence of motion, not only of projectiles, but also of rotating grindstones, bouncing balls and swinging bells. Buridan's consideration of *impetus* in relation to these everyday devices was in effect an articulation of a fairly well-developed construct (and an extension of it) into categories of motion not previously considered. In this manner, Buridan's bell is the ' proto-pendulum ', and his example is the initial recognition of pendular motion as a significant case.

Oresme's role marks the second stage of the recognition process. He transformed and vivified Buridan's original mode of expression first by linking the swinging object with an imaginary case of great intellectual charm, and second by stripping the bell of its concrete manifestations, leaving only its essential elements, a weight hanging from a cord. Buridan's ' proto-pendulum ' became a true pendulum with Oresme. We should note that although Oresme modified Buridan's theory of *impetus* in important ways elsewhere in his works, none of these purely theoretical alterations seems significant here; Oresme's transformations in this case pertain only to the exemplary material. The imaginary case of a weight falling within a trans-global perforation had, as we have seen, its origins as a means of exemplifying a commonplace of Peripatetic physics, the tendency of all earthly objects to move toward the center of a (spherical) universe. As this image was taken up by commentators after Adelard of Bath, it seems to have engendered certain ancillary problems, and the chronology of Oresme's writings indicates that he was interested in such problems even before he had absorbed the fuller implications of Buridan's *impetus* theory. In his mature works he effectively equated the two types of motion shown by the falling weight and the swinging stone, using the latter as a visible analogy for the former and presenting an *impetus*-based explanation for both.

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Oresme's role presents no special problems for analysis; Buridan's original act of recognition is still somewhat problematic, however. We know that he was deeply interested in and committed to the idea of *impetus*, and we can recognize in his extant writings a marked tendency to think in terms of commonplace, everyday objects. Within the limits presented by these facts, we should also place an important environmental factor. Jean Buridan lived in an urban society in which heavy, swinging bells had become a ubiquitous feature of daily life. In this he differed from virtually all his important philosophical predecessors; no Graeco–Roman thinker, no Chinese philosopher, nor any Islamic student of Peripatetic thought ever saw a heavy, swinging bell. It would seem therefore that the presence of such machines played a crucial part in the directions Buridan's thought took, for unlike any purely natural phenomenon, pendular motion could not exist until it had been brought into being through human artifice. Naturally, it is a ' simple ' matter to make a pendulum; equally obviously there were a few devices that could display pendular motion even in antiquity (plumb bobs, suspended battering rams, and swings). None of these ever became part of the mental baggage of natural philosophers, perhaps because they were rather uncommon devices which displayed pendular motion in transient fashions. The church bell as it had developed by Buridan's time was fully capable of being employed as a proximate physical model to suggest a certain type of motion in need of analysis. Buridan's choice of such a machine in his commentary was probably not fortuitous or trivial. In the virtual absence of any other candidates, what else could have served the bell's purpose? In place of the ' *Gestalt* switches ' or ' blurring of distinctions ', our analysis rests on the assertion that the pendulum first became visible when it was accidentally embodied in a machine intended for western Christian liturgical purposes. It was raised to the status of theoretical or philosophical significance when a particular thinker with a penchant for everyday examples grasped that the motion of a bell presented certain problems that could be treated within the framework of a conceptual modification of earlier physical theories. ' Knowing ', therefore, did not pre-determine ' seeing ' in this instance; rather what was known and what was seen both evolved over time independently until in the fourteenth century they were able to intersect each other in the thought of Buridan.

7. Epilogue: Oresme and Galileo

This study of the scholastic pendulum is meant to provide the necessary background for an understanding of how Galileo came to hold the doctrine of ' isochrony ', that is, the thesis that the swings of a pendulum take place in equal time even though the arcs of swing may differ greatly. Galileo's accomplishment is a separate tale to be told elsewhere in due course. It should be obvious that the scholastic pendulum was not in any significant way an anticipation of the Galilean thesis; nothing in either Buridan or Oresme (or any of their followers) suggests any concern with period, length of the pendulum, differences in weight, or any other important element of Galileo's pendulum. Indeed, the way in which Oresme formulated the pendulum almost completely removed it from the category of problem. His example was virtually closed, self-contained, and lacking any loose ends for further consideration. Perhaps for this reason, Oresme's example, especially the ' hole in the earth ' image,

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was reiterated without significant change from his day to Galileo's own time.[47] It even achieved a sort of secular immortality when Erasmus devoted one of his *Colloquies* to it;[48] Galileo used it himself in the *Dialogue concerning the two chief world systems*,[49]

In a curious way, however, the scholastic pendulum played a significant role in the discovery of isochrony, though its real importance is apparent only if one looks closely at the very earliest writings of Galileo, and particularly at the treatise he called *De motu* (about 1590). This was the great Florentine's first attempt at sweeping modifications of the Aristotelian-scholastic theories of local motion, and though the effort miscarried (Galileo never permitted its publication in his lifetime), it contains important germs of concepts he later developed in different ways. Among these is the pendulum, which Galileo employs in *De motu* in the scholastic fashion to illustrate the relationship between *impetus* and weight: a heavier pendulum remains in motion longer than a light pendulum.[50] For reasons that need not concern us here, Galileo abandoned the general line of argument advanced in *De motu* and went ahead in different ways to accomplish his major discoveries in mechanics. First among these was isochrony, which he announced in a letter written late in 1602.[51] This letter describes an experimental demonstration of his thesis, and there is a close relationship between this experiment and that contained in *De motu*. In the earlier work, of course, Galileo had included the pendulum to substantiate an established, unobjectionable scholastic fact; in the letter of 1602, the pendulum's properties have become an object of inquiry in their own right. Despite this important difference, there is ample evidence to suggest that Galileo discovered isochrony chiefly by performing the experiment with

[47] The major figure in passing on Oresme's image was Albert of Saxony, who seems to have included only the 'hole through the earth' example without the pendulum (see Clagett (footnote 25), 566 and 569 for passages from Albert's *Quaestiones super . . . de caelo*). A revival of interest in Oresme and Buridan at the University of Paris in the early years of the sixteenth century helped reintroduce the pendulum in its original form. Thus in the sixteenth-century literature, one sometimes encounters the problem presented in Albertian fashion (without pendulum) (see, for example, Nicolo Tartaglia's *Nova scientia* of 1537, quoted in Stillman Drake and I. E. Drabkin, *Mechanics in sixteenth-century Italy* (Madison: University of Wisconsin Press, 1969), 75–76). At other times, the problem is presented in Oresmian fashion (with pendulum), for example, in Francesco Maurolico's *Cosmographia* (Venice: 1543), fols. 15v–16r, or in the appendix to Giambattista Beneditti's *Diversarum speculationum mathematicarum et physicarum liber* (1585). (Regarding Maurolico, see Duhem, *Etudes* (footnote 25), vol. 3, 195–196; on Benedetti, see Drake and Drabkin *supra*, 235.)

[48] Colloquy 'Problemata' in Craig R. Thompson (trans.), *The colloquies of Erasmus* (Chicago: University of Chicago Press, 1965), 530 *ff*. Erasmus is patently dependent on Albert of Saxony for his knowledge of the subject.

[49] Ed. and trans. Stillman Drake (2nd ed.; Berkeley and Los Angeles: University of California Press, 1970), 22 and 227.

[50] For the passages in question see Galileo, *Opere* Ed. Naz., vol. 1 (Firenze: 1898), 335 and the memorandum note (p. 413). An English version of the main text of *De motu* appears in I. E. Drabkin and Stillman Drake (eds. and trans.), Galileo Galilei, *On motion and on mechanics* (Madison: University of Wisconsin Press, 1960), see p. 108. Galileo's memoranda to *De motu* are given in English in Stillman Drake and I. E. Drabkin, *Mechanics in sixteenth-century Italy* (Madison: University of Wisconsin Press, 1969); see p. 383.

[51] See footnote 2.

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two pendula which he had mentioned in *De motu* more than ten years before.[52] As noted above, a full discussion of Galileo's discovery must be reserved for a subsequent paper, but if the thesis here advanced is correct, then the scholastic pendulum as formulated by Oresme came to serve quite a different purpose than that for which it was originally intended in fourteenth-century Paris. Quite literally, it became one of the starting-points from which Galileo moved toward the creation of a new science of motion.

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[52] In the *De motu* passages, Galileo describes a thought experiment in which two cords of equal length have different weights attached to them. When started in motion simultaneously through the same arcs, the heavier of the two will remain in motion long after the lighter has come to rest. In the context of Galileo's argument at that time, the difference in duration of motion was all he needed to show. In the experiment described in 1602, two pendula of equal length whose weights have no special significance (they can be equal or not) are started simultaneously but through different arcs; Galileo claims that they will keep in step within a reasonable degree of accuracy regardless of the angular distance each traverses. Having established the synchrony of two pendula, Galileo can demonstrate on logical and mathematical grounds the isochrony of a single pendulum. For fuller discussion of this issue, see my forthcoming paper, 'Galileo's discovery of isochronous motion in the pendulum'.